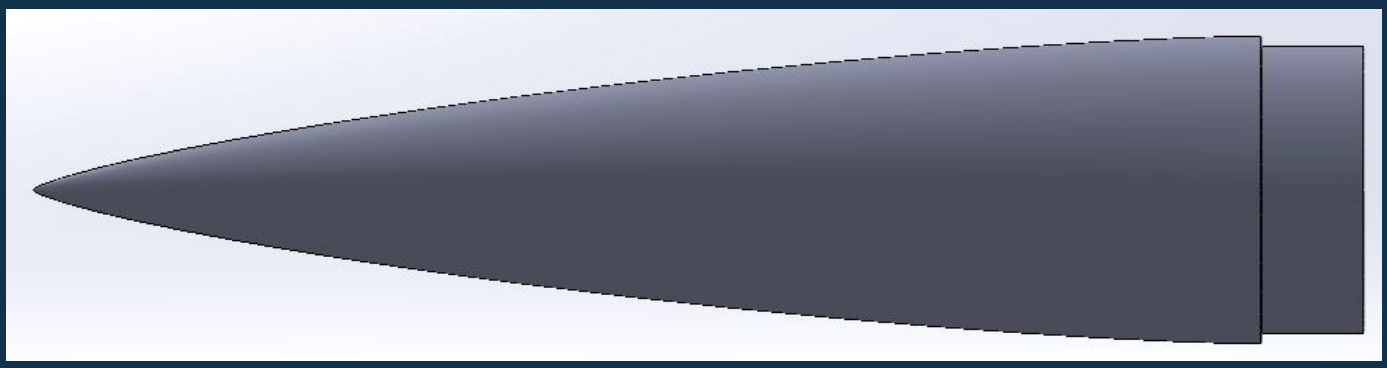
DAEDALUS

NOSE CONE + RECOVERY INTERFACE

Engineering reconstruction and design comparison



Original CDR nose CAD; clean side profile

MARCUS GREENAN | STRUCTURES LEAD

Native SolidWorks CAD and solved bearing-load studies

Original rounded eyelet reconstruction versus wider concept

Manufactured and ground-tested; never flight-tested

Evidence-based retrospective • 14 October 2026

PROJECT RECORD

What was built and what is known

Daedalus was a student rocket program with a 4,000 ft design target. The team manufactured and ground-tested the rocket. No flight test or measured ground-test values appear in the supplied record.

Structures Lead scope

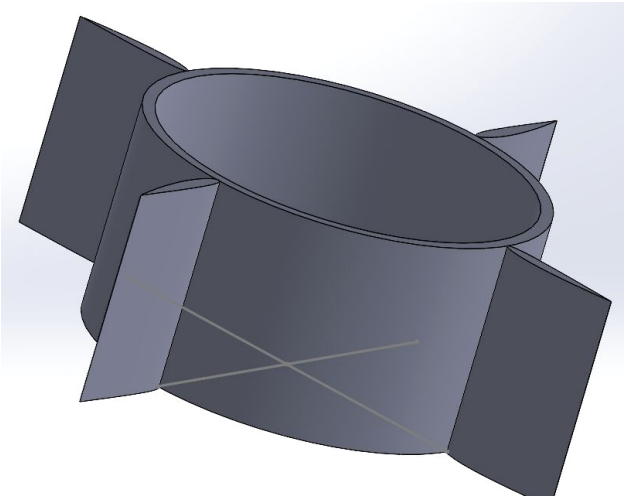
- Nose shape and body shoulder; recovery anchor load path and fit.
- Structural manufacturing support, shoulder/body fit checks and hardware integration.
- Retrospective work here: native CAD reconstruction, matched studies, and release criteria.

CDR dimensions and predicted performance

Quantity	CDR value	Evidence class
Overall envelope	25 in × 1.5 in	Documented design
Nose length	6 in Von Kármán	Documented design
Rocket mass	0.471 kg	CDR design input
Apogee	4,086 ft	Historical prediction
Static margin	1.23 calibers	CDR prediction



New eight-component SolidWorks envelope concept, not as-built geometry.



Historical four-fin can CAD from the source CDR.

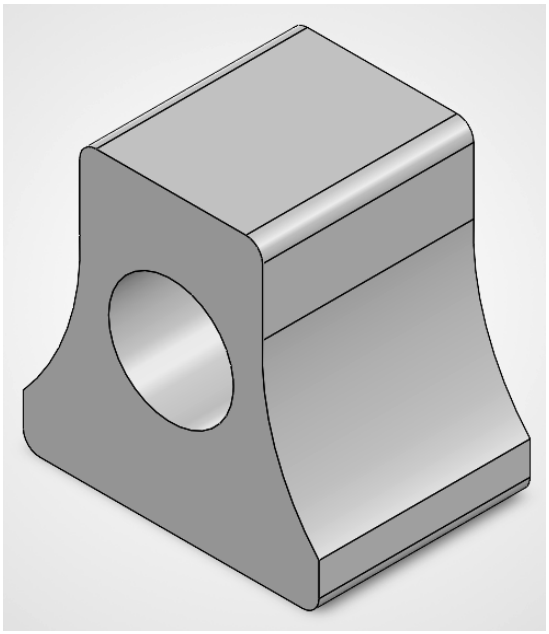
Source boundary: CDR dimensions, fin-can image, and performance prediction are historical. The eyelet hole, nose cavity, fin-can sleeve and assembly placements are reconstructed assumptions.

CDR pp. 7-12, 19-21; Native_SolidWorks/Rocket_Interference_Check_execution.log

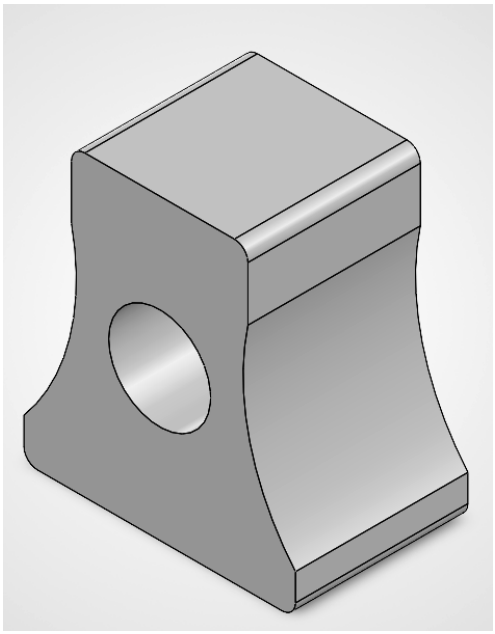
NATIVE CAD

Rounded eyelet reconstruction and redesign

The source image shows a curved eyelet, not a square tab. The reconstructed SolidWorks part follows its rounded, flared outline and preserves the documented 0.7 × 0.7 × 0.5 in envelope.



Original-style reconstruction: 17.78 × 17.78 × 12.7 mm.



Improved concept: 20 × 22 × 12.7 mm.

Feature	Original reconstruction	Improved concept
Hole diameter	7.5 mm assumed	7.5 mm assumed
Hole center height	9.5 mm assumed	11.0 mm assumed
Top ligament	4.53 mm computed	7.25 mm computed
Thickness	12.7 mm CDR envelope	12.7 mm retained
Geometry validation	1 solid; Check3 zero faults	1 solid; Check3 zero faults

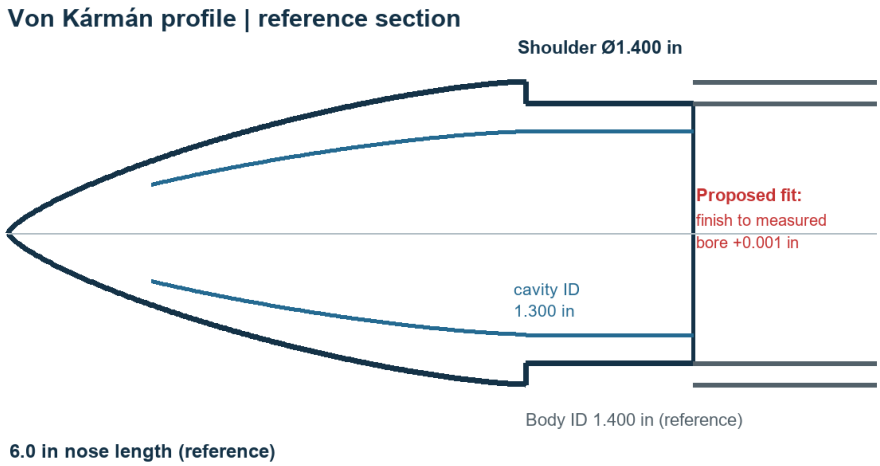
Engineering choice

Keep the hole and thickness fixed so the bearing interface and baseline section stay comparable. Add width and top ligament to spread load, accepting increased mass. The exact original hole, arcs, root, print direction, and attachment were not dimensioned in the CDR.

Original CAD image: 02_Original_CAD_Images/page08_img2_88.jpeg; native part verification logs.

Shoulder fit and nose retention

I sized a conditional friction-fit target from coast separation and commanded ejection, then specified a measurable pull-force acceptance window.



Original CDR dimensions, reconstructed section. The friction-fit target below has not been applied to native CAD.

From CDR zero clearance to initial interference

$$t_{r,\mathrm{ref}} = \frac{D_s - D_c}{2} = \frac{1.400 - 1.300}{2} = 0.050 \text{ in}$$

$$C_{\mathrm{ref}} = D_b - D_s = 1.400 - 1.400 = 0 \text{ in}$$

$$C_{\mathrm{fit}} = D_b - D_s = 1.400 - 1.401 = -0.001 \text{ in}$$

$$t_{r,\mathrm{target}} = \frac{1.401 - 1.300}{2} = 0.0505 \text{ in} = 1.283 \text{ mm}$$

Conditional flight and release window

$$F_{\mathrm{sep}} \le \frac{m_n C_D q A}{M_{\mathrm{coast}}} = \frac{(0.050)(1.0)(40.6)}{0.300} = 6.77 \text{ N}$$

$$15 \le F_{\mathrm{pull}} \le 20 \text{ N}; \quad \frac{15}{6.77} = 2.22$$

$$F_{\mathrm{eject,min}} \ge 35 \text{ N} > (20 + 0.49) \text{ N}$$

$$\Delta p_{\mathrm{min}} = \frac{35 \text{ N}}{9.93 \times 10^{-4} \text{ m}^2} = 35.2 \text{ kPa}$$

$$F_{\mathrm{pull}} \simeq \mu p \pi D L \quad (\mu, \text{p require measurement})$$

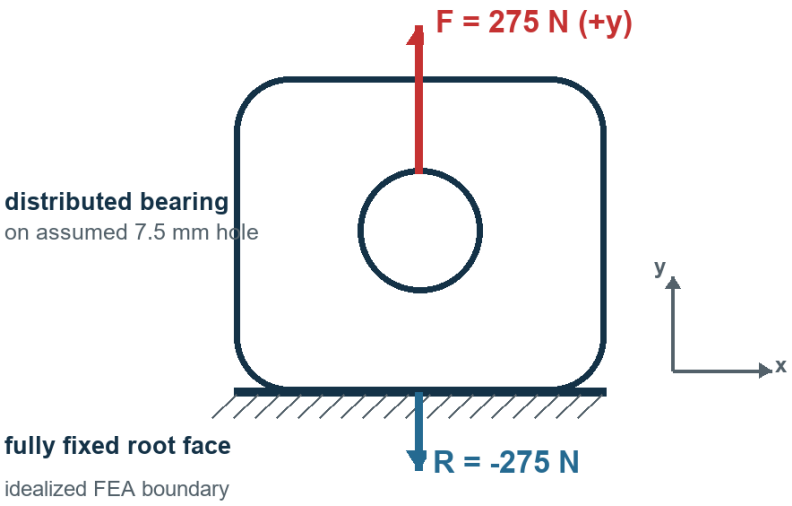
Assumptions: 241 m/s CDR predicted peak speed, sea-level air, drag coefficient 1.0, 0.300 kg coast mass, 0.050 kg nose mass. Coast drag deceleration gives a conservative 6.77 N separation screen with local nose drag neglected; powered acceleration seats the nose. Finish the shoulder near measured bore +0.001 in, then require 15–20 N measured pull and at least 35 N measured ejection force. Actual masses, drag, bore, friction, pressure and release dynamics need measurement.

A 0.001 in interference is only a finishing start point. Diameter cannot predict extraction force without surface, stiffness and friction data; flight hardware must pass repeated pull/ejection tests. The CDR wall sentence (0.100 in) conflicts with its diameters.

LOAD PATH

Recovery opening: conditional screening case

A hypothetical deployment 1.5 s after apogee supplies an explicit load basis. Opening shock and the added structural screening multiplier are separate assumptions. This does not predict the maximum actual cord load.



Idealized +y hole bearing and opposing fixed-root reaction; connection and line angle omitted.

$$A = \frac{\pi D^2}{4} = 0.2919\,\mathrm{m}^2$$

$$V \approx gt = (9.81)(1.5) = 14.715\,\mathrm{m\,s^{-1}}$$

$$F_{\mathrm{open}} = \tfrac{1}{2}\rho V^2 C_D A K_o = 135.5\,\mathrm{N}$$

$$F_{\mathrm{screen}} = K_s F_{\mathrm{open}} = 271\,\mathrm{N} \Rightarrow 275\,\mathrm{N}$$

Hypothetical delay	V ≈ gt	F_open	2 × F_open
1.0 s	9.81 m/s	60.2 N	120.4 N
1.5 s	14.715 m/s	135.5 N	271.0 N
2.0 s	19.62 m/s	240.9 N	481.7 N

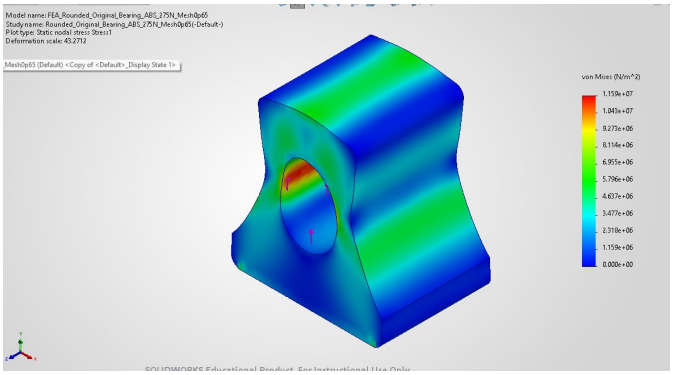
Assumed D = 0.6096 m, rho = 1.225 kg/m3, Cd = 1.75, opening multiplier = 2, screening multiplier = 2. Mass 0.471 kg is context, not an input to this drag equation. Inflation, line stretch, angle, motion and pressure history need measurement.

SOLIDWORKS SIMULATION

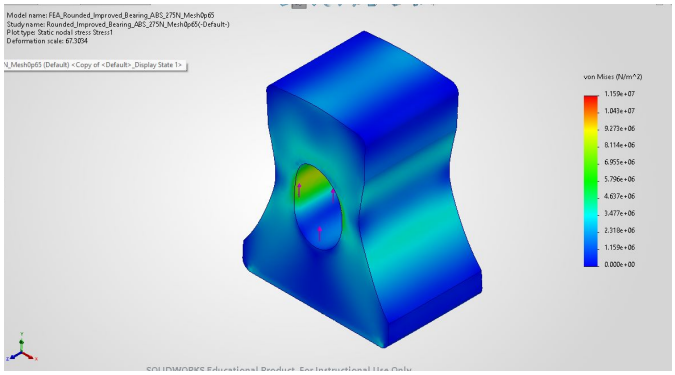
Matched bearing-load model

Both rounded designs use the same built-in isotropic ABS, load magnitude, hole diameter, thickness, root fixture and solver settings. Only the rounded outer profile changes.

Input	Saved study setting
Analysis	Linear static; high-quality solid elements
Load	275 N +Y sinusoidal bearing on cylindrical hole
Fixture	Entire bottom y = 0 face fully fixed
Coordinate system	Axis coaxial with 7.5 mm hole
Mesh verification	275 N: 0.65 mm; earlier 174.59 N: 1.20 to 0.65 mm
Outputs	von Mises, resultant displacement, fixture reaction



Actual original 275 N stress plot; 11.591 MPa peak.



Actual wider-concept 275 N plot; 8.869 MPa peak.

What this comparison isolates

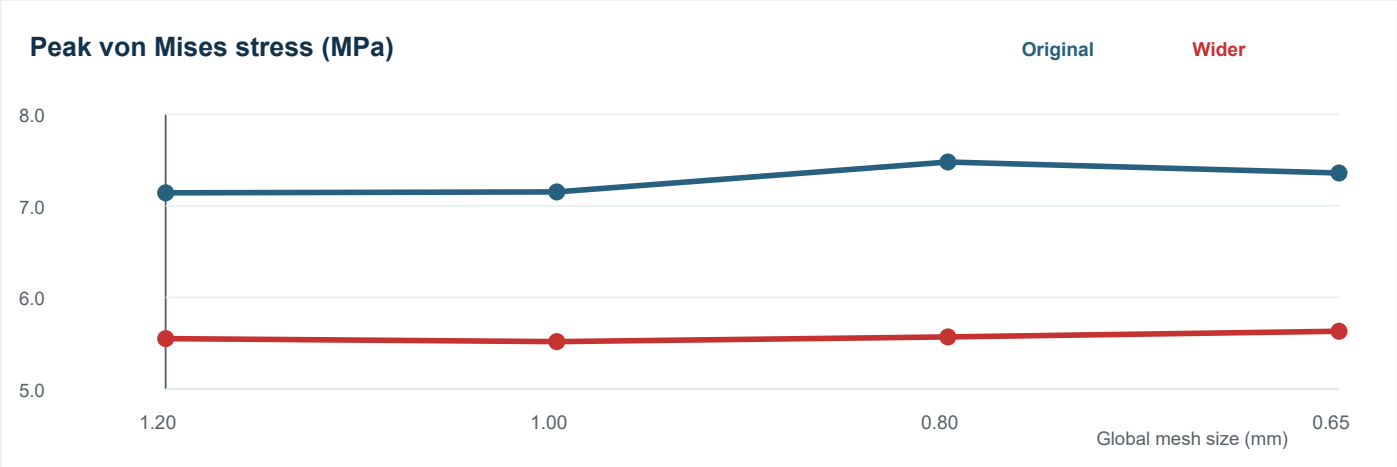
Hole, thickness, material, load and root fixture are held fixed; only the outer profile changes. At 275 N, the nominal net-section screen falls from 2.106 to 1.733 MPa, consistent with the solved trend. The fully fixed face still idealizes the real nose connection.

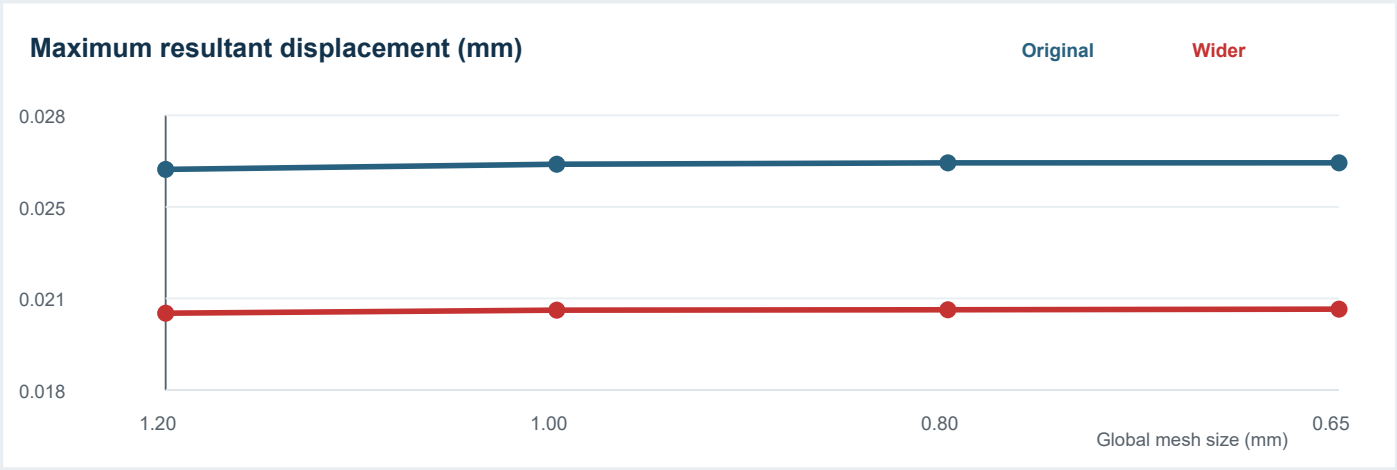
Both saved 275 N plots use the same 0–11.591 MPa color scale and isometric view. Original/improved displacement: 0.041379/0.032988 mm. Fixture reactions: −275.0017/−274.9988 N. Real pin/contact, line angle and root connection are omitted.

VERIFICATION

Mesh refinement and reaction equilibrium

Three refinements after the initial 1.20 mm mesh were solved on each design. The native solver files and extracted CSV remain in FINAL_SUBMISSION.





$$\frac{7.359 - 5.631}{7.359} \times 100\% = 23.5\%$$

$$\frac{0.026271 - 0.020944}{0.026271} \times 100\% = 20.3\%$$

Earlier 174.59 N mesh series: reactions within 0.023 N; 275 N final-mesh reactions within 0.002 N.

Peak stress is not perfectly monotonic with mesh; the comparison is a screening trend.

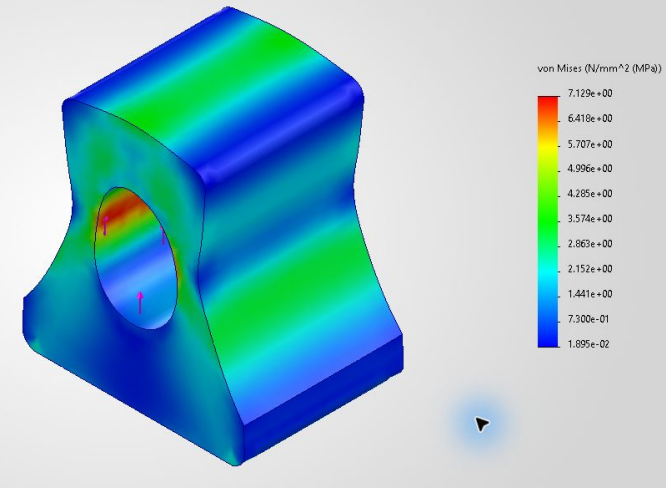
Graphs show the earlier 174.59 N mesh series. The 275 N final-mesh peaks follow the expected linear factor 275/174.5927 = 1.57509. This does not establish printed PLA strength or root-attachment safety.

MATERIAL SENSITIVITY

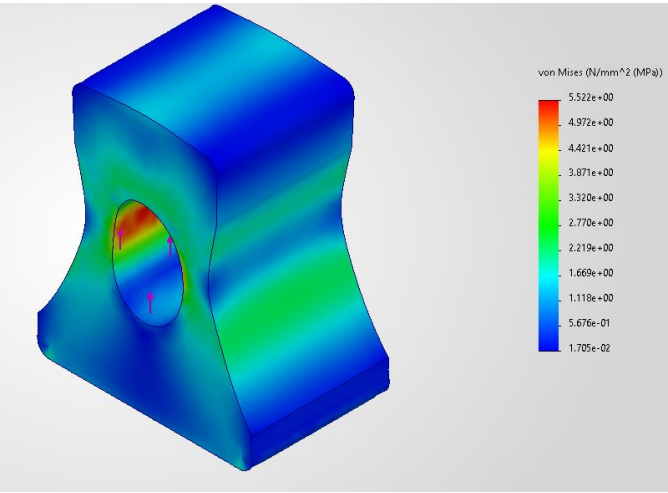
Assumed isotropic PLA is not a print allowable

A custom material study tests stiffness sensitivity only. PLA was entered with E = 3.0 GPa, v = 0.35, G = 1.111 GPa and density = 1240 kg/m³. No tensile or interlayer strength was entered.

Design / material	Peak MPa	Disp. mm	Reaction N
Original / ABS	7.143	0.02603	−174.596
Improved / ABS	5.550	0.02080	−174.576
Original / assumed PLA	7.129	0.01731	−174.581
Improved / assumed PLA	5.522	0.01382	−174.566



Saved original PLA assumption, 1.20 mm mesh.



Saved improved PLA assumption, 1.20 mm mesh.

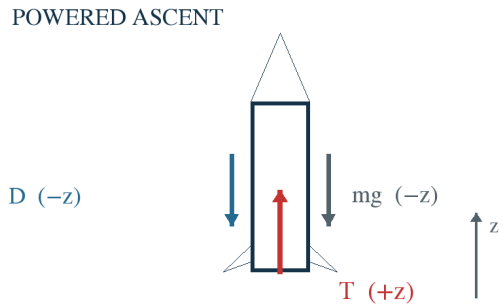
As expected for a linear, force-controlled elastic model, changing modulus has little effect on stress but changes displacement. These studies do not measure print orientation, contour/infill paths, bonding, porosity, defects, temperature dependence, or strength. No factor of safety is claimed.

What would turn sensitivity into validation

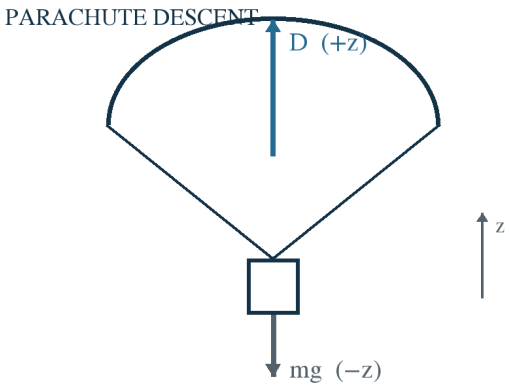
- Test coupons printed in the eyelet orientation for elastic modulus and tensile/interlayer strength.
- Pull representative eyelet-to-nose assemblies with the actual pin and line angle; record failure mode.
- Use measured properties and load envelopes before calculating a release factor of safety.

Assembly, ascent and recovery envelopes

The eight-component native assembly has no volumetric overlap in the nominal zero-coincidence interference check. Its fins, can sleeve and axial positions remain packaging assumptions; placements are transforms rather than verified mates.



October 2026 ascent free body; drag and mass variation remain assumptions.



October 2026 descent free body; chute coefficient is unknown.

First-principles screens using CDR inputs

$$a_{\text{mean}} \approx \frac{78 - (0.471)(9.81)}{0.471} = 155.8 \text{ m/s}^2$$

$$a_{\text{peak}} \approx \frac{108.5 - (0.471)(9.81)}{0.471} = 220.5 \text{ m/s}^2$$

$$SM = \frac{x_{CP} - x_{CG}}{D} = \frac{17.713 - 15.862}{1.500} = 1.234 \text{ calibers}$$

$$q = \tfrac{1}{2}\rho V^2 = \tfrac{1}{2}(1.225)(241)^2 = 35.6 \text{ kPa}$$

$$qA = (35.6 \text{ kPa}) \times \frac{\pi(0.0381 \text{ m})^2}{4} = 40.6 \text{ N}$$

$$A_{\text{chute}} = \frac{\pi(0.6096 \text{ m})^2}{4} = 0.292 \text{ m}^2$$

$$C_D = \frac{2mg}{\rho A v^2} \approx 2.52 \quad \text{(inferred from archived simulation)}$$

Acceleration neglects drag and mass loss; the dynamic-pressure screen assumes sea-level density. The chute drag coefficient is inferred from a historical descent prediction, not measured. Added eyelet mass needs a new CG and stability check.

No original OpenRocket file or installed simulation was available. Historical 4,086 ft apogee remains an unreproduced prediction, never a flight result.

CDR pp. 7-12, 19-21; Native_SolidWorks/Rocket_Interference_Check_execution.log

DECISION

What the evidence supports

Advance the wider rounded profile as a prototype candidate, not as a released recovery anchor. Its matched ABS model shows lower peak stress and displacement; the unmodeled connection and printed material behavior control the next decision.

Validation gates before release

- Measure the as-built shoulder/body dimensions and define acceptable retention and separation forces.
- Document the eyelet hole, radii, root geometry, print orientation, attachment and pin clearance.
- Test printed coupons and representative eyelet-to-nose assemblies in the actual build direction.
- Model the full lug/root connection under credible line directions and measured deployment loads.
- Assign measured masses and geometry, then rerun stability and OpenRocket performance.

Original-style reconstruction versus October 2026 concept

Engineering choice	Original-style	Wider concept
Outer envelope	17.78 × 17.78 × 12.7 mm	20 × 22 × 12.7 mm
Hole and thickness	7.5 mm assumed; 12.7 mm	Same hole and thickness
275 N net-section	2.106 MPa	1.733 MPa
275 N ABS peak	11.591 MPa	8.869 MPa

Lesson and interview defense

A wider section reduced stress in the matched conceptual model, but the printed eyelet-to-nose connection controls the release decision. Lead with the load path, show the conditional load and matched meshes, then explain why the result calls for a physical pull test rather than a claimed factor of safety.

Evidence in the editable package

Native_SolidWorks contains the parts, assembly, drawing, solved CWR files and material library. SolidWorks_Screenshots contains inspected geometry and plot images. The mesh, material and 275 N CSV files preserve solver outputs. The load calculation script and this Python file are editable sources.